

Thematic Analysis Report 1: Poster Motivation

Raw OCR poster text from Documents/OCR 1.1/Raw; authors and references were ignored.

Categories were defined after reviewing the full poster set so the analysis does not depend on poster order.

1) Category summary

Category	Definition	Count	Posters
Human disease models and translational health	Posters where a named human disease, disorder, or direct health model is the main reason for the work. Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.	11	Avetisyan et al. (2023) - CCC SP23.txt Brown et al. (2025) - CCC SP25.txt Cabral et al. (2025) - CCC FA25.txt Haubelt & Alcazar et al. (2025) - CCC FA25.txt Logan et al. (2025) - CCC FA25.txt Ojala et al. (2025) - LU SP25.txt Paderna et al. (2025) - CCC SP25.txt Paramo-Ojeda et al. (2025) - CCC SP25.txt Rodriguez (2025) - CCC SP25.txt Trevino et al. (2022) - CCC SP22.txt Zlaket et al. (2023) - CCC SP23.txt
Metabolism, digestion, and energy homeostasis	Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.	7	Berbouti et al. (2025) - CCC FA25.txt Camara et al. (2025) - LU SP25.txt Dehuelbes et al. (2025) - LU SP5.txt Lemus et al. (2025) - FA25 CCC.txt Nii et al. (2025) - CCC FA25.txt Pedireddi et al. (2025) - CCC SP25.txt Tuttle et al. (2025) - CCC SP25.txt
Neurobiology, behavior, and signaling	Posters centered on nervous-system signaling, sleep, feeding behavior, gut-brain communication, or related signaling pathways without a named disease being the main driver.	3	Gill & Alcazar (2025) - CCC FA25.txt Godoy-Pena et al. (2025) - CCC FA25.txt Luera et al. (2025) - CCC FA25.txt
Basic gene regulation, development, and cell biology	Posters mainly about gene function, patterning, pigmentation, transcriptional control, or cell biology. These tend to be foundational biology studies with only a secondary disease connection, if any.	3	Holmes (2025) - CCC SP25.txt Meraz et al. (2023) - CCC SP23.txt Sakana et al. (2025) - CCC SP25.txt
Defense, detoxification, and stress response	Posters focused on immune defense, phagocytosis, detoxification, or environmental resistance. Examples include immune pathways, phagosome acidification, and insecticide resistance.	3	Diaz et al. (2025) - CCC SP25.txt Ford et al. (2023) - CCC SP23.txt Henriquez et al. (2025) - COD WI24.txt
Other	Posters that do not fit the five main categories above.	0	—

2) Poster-by-poster evidence

Avetisyan et al. (2023) - CCC SP23.txt — Expression Analysis of Autism Related Genes Across the D. Melanogaster Midgut

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: Primary motivation is autism-related gene expression in the gut-brain axis.

Evidence from the poster text:

bullet These genes, which play a role in the upkeep of the gut-microbiome-brain axis, have been identified as autism related.

bullet These genes such as these are being studied to further research autism, and how treatment of the midgut can affect autism management.

bullet Our research of autism related genes we found that those genes which are expressed more evenly between the CNS and midgut have similar trends when plotted across all regions of the midgut.

Berbouti et al. (2025) - CCC FA25.txt — Trehalase Expression in the Drosophila Midgut: Differential Expression across Anterior and Posterior regions Dy

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Carbohydrate metabolism and regional digestion are the main themes.

Evidence from the poster text:

bullet Trehalose is a critical sugar in Drosophila melanogaster, in which its breakdown by trehalase provides glucose for energy metabolism.

bullet The pathway of trehalose catabolism involves two transporters (TRET1/GLUT1) and one (TREH) enzyme.

bullet These results indicate that, contrary to our hypothesis, expression of trehalose catabolism genes is regionally specialized rather than uniform.

Brown et al. (2025) - CCC SP25.txt — Analysis Of Gene Expression Of Acids Reflux Related Genes Across The Midgut

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A disease mechanism study, but the poster still focuses on midgut gene expression.

Evidence from the poster text:

bullet This study explores gene expression related to acid reflux-associated pathways in the Drosophila melanogaster midgut, it focus on acidification processes that influence metabolic gene regulation.

bullet Fundamental to this investigation is the gene vha16-1 (FBgn0262736) with the human gene ortholog being ATP6VOC, which encodes a subunit of the vacuolar H⁺-ATPase (V-ATPase), necessary for acidifying the midgut—especially within the copper (Cu) cell region.

bullet This localized acidification plays a vital role in digestion and maintaining metabolic homeostasis.

Cabral et al. (2025) - CCC FA25.txt — [title not found in OCR]

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: The disease link is central, even though the poster also contains basic brain-expression work.

Evidence from the poster text:

bullet The methyl-CpG binding protein 2 (MECP2) gene is essential for neuronal nervous system.

bullet Mutations in MECP2 are linked to Rett syndrome and related neurodevelopmental disorders; however, the characterized.

bullet The results indicate that MECP2 is broadly expressed throughout the mouse brain, with notably higher expression in the cortex and hippocampus compared to subcortical regions.

Camara et al. (2025) - LU SP25.txt — Functional Difference Between Amy-D, Amy-P and Amyrel in Drosophila Midgut

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: The main motivation is amylase function and digestion; disease examples are secondary context.

Evidence from the poster text:

- * We studied expression of three fly amylase genes: Amy-d, Amy-p, and Amyrel.
- * These are orthologs to human amylases and may relate to diseases like diabetes and Alzheimer's (Leue, 2016).
- * Amyrel is functionally distinct, indicating a different digestive role and possibly stress-linked regulation.

Dehuelbes et al. (2025) - LU SP5.txt — Regional Expression of Key Metabolic genes in Drosophila Midgut: CRAT, MANF, NPY and their relation to each other R

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Metabolism and energy balance are the dominant themes.

Evidence from the poster text:

- * Acetyl-CoA links carbohydrate, fat, & protein metabolism * Energy Production: Enters the Krebs Cycle to produce ATP * Precursor for synthesis of fatty acids, cholesterol, and acetylcholine * Carnitine O-Acetyltransferase (CRAT) is a mitochondrial enzyme involved in fatty acid metabolism * Regulates acetyl-CoA levels and mitochondrial energy balance
- * Plays a role in cellular energy homeostasis and may influence feeding behavior * MANF Promotes neuron survival, regulates ER stress response, supports protein folding; involved in cellular stress responses and have a coordinated role in cellular energy * NPY Controls appetite, energy balance, circadian rhythms, and stress response.
- * Shared role in regulating metabolic adaptation and energy homeostasis * The Drosophila gut serves as a valuable model for studying the CRAT gene due to its conserved metabolic pathways
- * Therapeutic Exploration: Explore CRAT as a potential target for treating metabolic disorders in humans.

Diaz et al. (2025) - CCC SP25.txt — V-ATPase Gene Expression Reveals Cellular Defense Zones in the Copper Region of the Drosophila Viidgut »

Assigned category: Defense, detoxification, and stress response

Definition: Posters focused on immune defense, phagocytosis, detoxification, or environmental resistance. Examples include immune pathways, phagosome acidification, and insecticide resistance.

Note: This is a cellular defense / immunity poster, not a named disease poster.

Evidence from the poster text:

- * To explore how different regions of the Drosophila melanogaster midgut contribute to immune defense, we examined the regional expression of V-ATPase genes involved in phagosome acidification.
- * Phagocytes (cells that undergo phagocytosis) engulf and digest harmful debris into the cell.
- * If this system fails, it can lead to infections, disease, or inflammation — making phagocytosis an important process for helping protect the body.

Ford et al. (2023) - CCC SP23.txt — Cytochrome P450 genes providing insecticide resistance found in al, p1 and p2 4 midgut regions of Drosophila

Assigned category: Defense, detoxification, and stress response

Definition: Posters focused on immune defense, phagocytosis, detoxification, or environmental resistance. Examples include immune pathways, phagosome acidification, and insecticide resistance.

Note: Environmental defense and xenobiotic resistance are the main motivations.

Evidence from the poster text:

- * Some Cytochrome P450 genes play an important role in the development of insecticide resistance in Drosophila Melanogaster.
- * DDT is used as the insecticide of interest as it is a known to affect insects through the midgut.
- * Using the FlyBase database and DESeq2, we examined several different genes from the Cytochrome P450 gene family that were known to have roles in insecticide resistance in the al, pl and p2_4 midgut sections.

Gill & Alcazar (2025) - CCC FA25.txt — Differential Expression of Serotonin Receptor Genes Htr1a and Htr2a in the Prefrontal Cortex and Striatum

Assigned category: Neurobiology, behavior, and signaling

Definition: Posters centered on nervous-system signaling, sleep, feeding behavior, gut-brain communication, or related signaling pathways without a named disease being the main driver.

Note: A basic neuroscience signaling study.

Evidence from the poster text:

- * In this poster, I examined two serotonin receptor genes, those being Htr1a and Hir2a to analyze how strongly expressed they were between the prefrontal cortex and the striatum in mice using RNA-seq data.
- * The purpose of this study was to test if serotonin receptor signaling is higher in the PFC compared to the STR as hypothesized.

bullet is the prefrontal cortex and the striatum In the brain the role of the prefrontal cortex is decision making and emotional regulation (5).

Godoy-Pena et al. (2025) - CCC FA25.txt — Regulation of Sleep Quality: Analysis of Gene Expression Using Bulk RNA-seq Data

Assigned category: Neurobiology, behavior, and signaling

Definition: Posters centered on nervous-system signaling, sleep, feeding behavior, gut-brain communication, or related signaling pathways without a named disease being the main driver.

Note: Sleep and behavior regulation are the key ideas.

Evidence from the poster text:

bullet The main purpose of this study was to determine what genes dictate the kind of sleep drosophila flies get, and connect it to a gut-brain region.

bullet We first searched for a wide range of genes that were known to influence the drosophila fly's circadian rhythm.

bullet The genes CCHa1-R/CCHa2-R was already well researched and well known for the effects it had on the drosophila flies quantity and quality of sleep, so we decided to use that as well as the gene NPF for further investigation.

Haubelt & Alcazar et al. (2025) - CCC FA25.txt — Exploring DRD2: The Association of PD Risk with Dopamine Receptors in the Dopaminergic Synapse

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: Parkinson's disease is the organizing theme of the poster.

Evidence from the poster text:

bullet I was interested in the genes, pathways, and mechanisms related to the neuropsychiatric disorder Parkinson's Disease and wanted to investigate the specific genes involved in the dopaminergic synapse.

bullet A normal variants of the Drd2 gene has been implicated in many neurological disorders including Parkinson's Disease in humans (Noble, E).

bullet The dopaminergic synapse involves genes, including tyrosine hydroxylase (Th), catechol-O-methyltransferase (Comt), dopamine receptor D2 (Drd2), and dopa decarboxylase (Ddc).

Henriquez et al. (2025) - COD WI24.txt — 6 Immune Pathways Exhibit Similar Patterns of Gene Expressions in Drosophila melanogaster Midgut

Assigned category: Defense, detoxification, and stress response

Definition: Posters focused on immune defense, phagocytosis, detoxification, or environmental resistance. Examples include immune pathways, phagosome acidification, and insecticide resistance.

Note: Immune-pathway activity is the core motivation.

Evidence from the poster text:

bullet Measurement of gene expression in six immune pathways (Toll, JAK/STAT, IMD, CGAS-Sting, TNF alpha, Hippo) suggests similar levels of immune pathway protein expression in the 10 midgut regions of Drosophila melanogaster characterized by Marianes and Spradling.

bullet The data indicates the copper and posterior regions are the most immunologically active suggesting a need for additional immune response in these regions.

bullet This may be due to bacterial invasion, pH levels, or other unidentified factors.

Holmes (2025) - CCC SP25.txt — Melanogenesis within the Drosophila Midgut: Neuromelanin and Dopamine

Assigned category: Basic gene regulation, development, and cell biology

Definition: Posters mainly about gene function, patterning, pigmentation, transcriptional control, or cell biology. These tend to be foundational biology studies with only a secondary disease connection, if any.

Note: Primarily pigmentation and dopamine biology, with a secondary neurodegenerative connection.

Evidence from the poster text:

bullet This research was focused in genes connected to melanogenesis (synthesis of melanin) and its antioxidant uses.

bullet It more specifically deals with melanogenesis within the midgut of Drosophila and the connection to melanogenesis in humans.

bullet Within Drosophila a gene that regulates its mating behaviors and pigmentation is known as "Y™" or "yellow".

Lemus et al. (2025) - FA25 CCC.txt — Macromolecule Breakdown and Absorption: Analysis of Differential Expression Using Single Cell and Bulk RNA-seq Data

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Digestive carbohydrate processing is the dominant theme.

Evidence from the poster text:

- bullet Productive macromolecule breakdown relies on certain genes expressed in specific regions of the *Drosophila* midgut.
- bullet Understanding differential gene expression in particular regions can uncover processes relevant to human digestion as well as metabolic disorders.
- bullet Our research aimed to compare the regional expression patterns of two genes both involved in starch and sucrose metabolism using single-cell as well as bulk RNA-sequencing datasets.

Logan et al. (2025) - CCC FA25.txt — [title not found in OCR]

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A translational cardiovascular model built around ACE/ANCE genes.

Evidence from the poster text:

- bullet In this project we focused on the *Drosophila Melanogaster* ANCE gene the fly ortholog of the human ACE gene.
- bullet Cardiovascular disorders are a prominent and widespread disease caused by genetic predispositions and environmental conditions.
- bullet The ACE gene found within humans is an angiotensin converting enzyme, which is a critical regulator of blood pressure and fluid balance.

Luera et al. (2025) - CCC FA25.txt — Region-Specific NPF/NPFR Signaling in the *Drosophila* Midgut

Assigned category: Neurobiology, behavior, and signaling

Definition: Posters centered on nervous-system signaling, sleep, feeding behavior, gut-brain communication, or related signaling pathways without a named disease being the main driver.

Note: Neuropeptide signaling and feeding behavior are the main ideas.

Evidence from the poster text:

- bullet We analyzed Neuropeptide F (NPF) and its receptor NPFR in *Drosophila melanogaster* to explore their role in gut-mediated feeding behavior. Using regional and single-cell RNA-seq datasets, we found that NPF is highest in the Copper (Cu) region and second-highest in the Iron (Fe) region, while NPFR is homeostatic.
- bullet These results support a model in which the brain-gut axis modulates feeding through region-specific NPF/NPFR interactions.
- bullet Neuropeptide F (NPF) is a conserved neuropeptide that regulates feeding, stress, and sleep behaviors in *Drosophila melanogaster* and is analogous to mammalian neuropeptide Y (NPY).

Meraz et al. (2023) - CCC SP23.txt — Tyrosine Kinase Expression in the Midgut Suggests a Role in the Copper Region

Assigned category: Basic gene regulation, development, and cell biology

Definition: Posters mainly about gene function, patterning, pigmentation, transcriptional control, or cell biology. These tend to be foundational biology studies with only a secondary disease connection, if any.

Note: This is mostly a gene-family / function poster with only a light cancer hint.

Evidence from the poster text:

- bullet Tyrosine kinases are enzymes related to cell growth and development.
- bullet The FRK gene, which is the human ortholog of Src42A, is involved in tumor-suppression and is localized to the small intestine.
- bullet We expected the FRK gene, along with the others, to be equally expressed across the *Drosophila* midgut because we assumed that all subregions of the midgut partake in tumor-suppression.

Nii et al. (2025) - CCC FA25.txt — Iron-Induced Modulation of Acetyl-CoA Carboxylase (ACC) in *Drosophila Melanogaster* : Impacts on Lipid Metabolism and Energy Storage

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Metabolic regulation and energy storage dominate.

Evidence from the poster text:

- Acetyl-CoA carboxylase (ACC) is the rate limiting enzyme that converts acetyl-CoA to malonyl-CoA, committing cells to fatty acid synthesis.

- We hypothesize that excess iron activates acetyl-CoA carboxylase, increasing malonyl-CoA production and driving fatty acid synthesis. To determine whether iron influences this pathway, we analyzed regional midgut expression of ACC in *Drosophila melanogaster*, an ideal model organism for studying evolutionarily conserved metabolic responses. Using RNA-seq data, FlyBase ortholog MEdp10g, 2nd segment specific expression analysis, we tested whether iron exposure induces lipogenic remodeling consistent with patterns observed in mammalian ACC. ACC showed 2 strong, region-specific increases in the 2nd segment, while the comparison gene ATP citrate lyase (ATPCL) exhibited relatively uniform expression, indicating that iron does not globally elevate all lipid-related enzymes.

- This contrast suggests that iron selectively enhances ACC activity rather than globally elevating all lipid-related enzymes.

Otala et al. (2025) - LU SP25.txt — Fly Gut is a good model for fatty acid metabolism disorders

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A clear inherited metabolic-disorder framing.

Evidence from the poster text:

- Related disorders and symptoms include: MCADD and Carnitine transporter deficiency, hypoglycemia, liver dysfunction, and muscle weakness.

Paderna et al. (2025) - CCC SP25.txt — Albinism and Melanogenesis Pathway: Analysis of Differential Gene Expression

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A pigmentation disorder with a direct human-disease connection.

Evidence from the poster text:

- In humans, the gene SLC45A2 is associated with oculocutaneous albinism type 4, a condition that affects the skin, eyes, and hair (Le et al).

- We hypothesized that the Mcol gene was related to the production of melanin; results were negative.

- In humans, oculocutaneous albinism type 4 (OCA4) manifests as reduced pigment in the skin, eyes, and hair (Hayashi).

Paramo-Ojeda et al. (2025) - CCC SP25.txt — Ras85D Expression in Drosophila Midgut Highlights a ~ Model for KRAS-Driven Colon Cancer

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: Cancer is the central motivation.

Evidence from the poster text:

- KRAS mutations drive many human cancers by promoting uncontrolled cell growth.

- The KRAS gene encodes a protein that controls cell growth signals.

- Cancer is a complex disease driven by mutations in genes that regulate cell growth, survival, and DNA repair.

Pedireddi et al. (2025) - CCC SP25.txt — Beyond the Anterior: Amylase Gene Expression Suggests Widespread Polysaccharide Digestion in the Drosophila Midgut |

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Carbohydrate digestion and absorption are the focus.

Evidence from the poster text:

bullet The A region has been found to have been responsible for the breakdown of polysaccharides that have been consumed by the fly, such as starch and sucrose, and the P region was also responsible for nutrient absorption (Marianes, 2013).
bullet In the dot plot of differentially expressed genes (Figure 1) between the A and P regions of the midgut, we delved deeper to investigate the genes related to starch and sucrose metabolism since it would relate to sugar metabolism.

Rodriguez (2025) - CCC SP25.txt — Differential Gene Expression of Galm1 in the Drosophila Midgut

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A direct metabolic-disorder model.

Evidence from the poster text:

bullet Mutations in the protein galactose mutarotase, or GALM, are associated with the disease galactosemia, which prevents the body from properly metabolising the sugar galactose.

bullet Galactosemia is a rare metabolic disorder that results from deficiencies in the GALM gene.

bullet This gene encodes for the protein galactose mutarotase, which catalyzes the interconversion of beta-d-galactose and alpha-d-galactose.

Sakana et al. (2025) - CCC SP25.txt — Differential Expression of the TALE Homeobox Genes in Drosophila Midgut patterning

Assigned category: Basic gene regulation, development, and cell biology

Definition: Posters mainly about gene function, patterning, pigmentation, transcriptional control, or cell biology. These tend to be foundational biology studies with only a secondary disease connection, if any.

Note: Developmental gene regulation and midgut patterning are the main goals.

Evidence from the poster text:

bullet Our results may help shed light on possible roles and hierarchy of how the TALE genes control patterning in the midgut.

bullet Gene expression patterns that control development and differentiation are often orchestrated by transcription factors.

bullet Homeothorax (hth) is a homeobox gene in Drosophila melanogaster known for its role in tissue patterning, stem cell maintenance, and organ development (Mann and Affolter, 2002).

Trevino et al. (2022) - CCC SP22.txt — 'Drosophila Melanogaster a Good Model System of Zellweger Spectrum Disorder

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: A classic disease-model poster built around peroxisomal biology.

Evidence from the poster text:

bullet Zellweger Spectrum Disorder (ZSD) is a genetic disorder that is caused by the mutation of 1 out of 13 different genes involved in the formation and function of peroxisomes.

bullet Understanding the disorder is important to the individuals affected and Drosophila can be a great tool to doing so.

bullet ZSD is a rare inherited disorder characterized by the absence/reduction of functional peroxisomes in cells, which are essential for beta-oxidation of very-long-chain fatty acids (Elumalai, 2021).

Tuttle et al. (2025) - CCC SP25.txt — Gene Expression Patterns in the Tryptophan-Kynurenine Pathway in Drosophila Melanogaster

Assigned category: Metabolism, digestion, and energy homeostasis

Definition: Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance. Examples include trehalase, amylase, fatty acid metabolism, galactose metabolism, and carbohydrate digestion.

Note: Metabolic pathway analysis with broader physiological implications.

Evidence from the poster text:

bullet The genes white and vermilion are essential for tryptophan metabolism, as well as the pigmentation found in the eyes of Drosophila.

bullet White affects the uptake of compounds that use tryptophan as a precursor, and vermilion is directly involved in the tryptophan to kynurenine pathway.

bullet We also analyzed the genes Speck (Fig 1.3) and Ddc (Fig 1.4) for their roles in tryptophan metabolism.

Zlaket et al. (2023) - CCC SP23.txt — ScpX: A Peroxisomal Gene and its Distinctive Paralogs

Assigned category: Human disease models and translational health

Definition: Posters where a named human disease, disorder, or direct health model is the main reason for the work.

Examples include autism, Rett syndrome, Parkinson's disease, cardiovascular disease, galactosemia, acid reflux, colon cancer, albinism, and Zellweger spectrum disorder.

Note: Peroxisomal disease biology is central.

Evidence from the poster text:

bullet In humans, the SCP2 gene is bifunctional as it expresses SCPx, functioning in | oxidation of fatty acid chains within peroxisomes, and SCP2, a sterol and glucose transporter (Fig.1).

bullet Errors within this gene can cause Zellweger's Syndrome, a buildup in several fatty acids like α -linoleic acid and arachidonic acid in the liver and brain, and issues in cholesterol uptake (Ezzat, 2022).

bullet Moreover, in *Drosophila M.*, all were located in peroxisomes, but only CG17597 shared cholesterol transport function.

Other category

No posters were assigned to Other; every poster fit one of the five main categories.